



COSMO INSTRUMENTS (I) PVT. LTD.

SPECIALIST IN AIR LEAK AND FLOW TEST

DIGITAL FLOW METER

MODEL DF-2820

ERROR AND TROUBLESHOOTING MANUAL

1. ERRORS

The tester indicates errors by displaying error messages on the LCD display. If a failure occurs during power-on sequence, measurement, calibration, or compensation sampling, the LEDs blink. In addition, the tester also outputs error signals.

1.1 Error Messages Displayed at Power-on and Treatments

1. ROM and RAM checks
If ROM or RAM becomes defective, the Hi and LO LEDs light up and all operations are disabled.
Contact Cosmo for repair.
2. Error # 5 **“FS Offset Exceeding Limit”** Out of flow sensor offset tolerance limits (beyond $\pm 10\%$ F.S.)
The flow sensor offset is abnormal. Contact Cosmo for repair. Note that entering the FLOW SENSOR OFFSET screen in CALIB mode temporarily cancels this error.
3. Error # 6 **“PS Offset Exceeding Limit”** Out of P sensor offset tolerance limits (beyond $\pm 2\%$ F.S.)
The P sensor offset is abnormal. Contact Cosmo for repair.
Inputting STOP temporarily cancels the error.

1.2 Error Messages and Treatments

ERROR#	Message	Description	Probable Cause and Treatments
2	A/D Converter Malfunction!	A/D converter malfunction	Replace the A/D converter. Contact Cosmo for repair.
5	FS Offset Exceeding Limit	FS offset tolerance limit has been exceeded (beyond $\pm 10\%$ FS)	The FS offset exceeds the tolerance limits. Contact Cosmo for repair. Note that entering the FLOW SENSOR screen in the CALIB mode temporarily cancels this error.
6	PS Offset Exceeding Limit	PS offset tolerance limit has been exceeded (beyond $\pm 2\%$ F.S.)	The PS offset exceeds the tolerance limits. Contact Cosmo for repair. Inputting STOP temporarily cancels the error.
8	Improper Test Pressure	Test pressure drop	Test pressure falls outside the set limits during measurement. The probable cause is an improper P.Lo (P.Hi for vacuum model) test pressure limit setting or P sensor failure.
10	PS output saturated	Saturated test pressure sensor output	The P sensor output during measurement has exceeded the tolerance limits. Check the PS offset and test pressure setting.
14	dT Sensor Over flow	Temperature sensor error	Contact failure of the temperature sensor. Check the connection. Replace the temperature sensor.
18	RTC Battery Discharge	Low battery voltage	The built-in battery is weak. Check the battery voltage and, if necessary, contact Cosmo for repair.